Mastering Quantum Mechanics — Problem 9.10

Matrices for $\hat{x}$ and $\hat{p}$ in the harmonic oscillator

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Problem 9.10

Matrices for $\hat{x}$ and $\hat{p}$ in the harmonic oscillator. With $X_{mn} \equiv \langle \varphi_m, \hat{x}\varphi_n \rangle$ and $P_{mn} \equiv \langle \varphi_m, \hat{p}\varphi_n \rangle$ in the energy basis $\{\varphi_0, \varphi_1, \dots\}$: (1) compute X_{mn}, P_{mn} for arbitrary m, n; (2) write the 3×3 truncations and classify them (Hermitian / symmetric / antisymmetric); (3) compute $[X, P]$ of the 3×3 matrices and comment.

Hints & Orientation

- **(1)** Expand $\hat{x}, \hat{p}$ in ladder operators (Eq. (9.3.12)) and act with $\hat{a}\varphi_n = \sqrt{n}\,\varphi_{n-1}$, $\hat{a}^\dagger\varphi_n = \sqrt{n+1}\,\varphi_{n+1}$ (Eq. (9.4.26)). Orthonormality $\langle \varphi_m, \varphi_n \rangle = \delta_{mn}$ then forces $m = n \pm 1$ — both matrices are *tridiagonal* with zero diagonal.
- **(2)** Just read off entries for $m, n \in \{0, 1, 2\}$. X is real symmetric (hence Hermitian); P is purely imaginary and Hermitian (equivalently $P^T = -P$).
- **(3)** Compute $[X, P] = XP - PX$. It is *not* $i\hbar\mathbb{1}$: a finite commutator is traceless, while $i\hbar\mathbb{1}$ is not. The mismatch sits in the bottom corner, where truncation severed the coupling to φ_3.

Solution

(1) General matrix elements

From Eq. (9.3.12), $\hat{x} = \frac{L_0}{\sqrt{2}}(\hat{a} + \hat{a}^\dagger)$ and $\hat{p} = \frac{i}{\sqrt{2}}\frac{\hbar}{L_0}(\hat{a}^\dagger - \hat{a})$ with $L_0 = \sqrt{\hbar/m\omega}$. Acting with Eq. (9.4.26) and projecting,

$$\boxed{X_{mn} = \sqrt{\frac{\hbar}{2m\omega}}\Big(\sqrt{n}\,\delta_{m,n-1} + \sqrt{n+1}\,\delta_{m,n+1}\Big)}, \tag{1}$$

$$\boxed{P_{mn} = i\sqrt{\frac{m\omega\hbar}{2}}\Big(\sqrt{n+1}\,\delta_{m,n+1} - \sqrt{n}\,\delta_{m,n-1}\Big)}. \tag{2}$$

(2) 3×3 truncation

Write $c \equiv \sqrt{\hbar/(2m\omega)}$ and $d \equiv \sqrt{m\omega\hbar/2}$. For $m, n \in \{0, 1, 2\}$,

$$X = c\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & \sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix}, \qquad P = i\,d\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -\sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix}. \tag{3}$$

X is real and symmetric, hence **Hermitian** ($X^\dagger = X,\ X^T = X$). P is purely imaginary with $P^\dagger = P$, so it is **Hermitian**; moreover $P^T = -P$, so it is also **antisymmetric**.

(3) Commutator

Using $cd = \sqrt{\frac{\hbar}{2m\omega}}\sqrt{\frac{m\omega\hbar}{2}} = \frac{\hbar}{2}$, a direct multiplication gives

$$\boxed{[X, P] = XP - PX = i\hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}}. \tag{4}$$

Comment. In the full (infinite-dimensional) space the canonical relation $[\hat{x}, \hat{p}] = i\hbar$ gives $[X, P] = i\hbar\mathbb{1}$. The truncated matrices reproduce this in the upper block ($i\hbar$ on the first two diagonal entries) but fail in the last entry, which is $-2i\hbar$ instead of $i\hbar$. The reason is structural: $\hat{x}$ and $\hat{p}$ connect φ_2 to φ_3, and dropping φ_3 removes the term that would have fixed the corner. Indeed a finite-dimensional commutator is always traceless, $\mathrm{tr}[X, P] = 0$ (here $i\hbar(1+1-2) = 0$), whereas $\mathrm{tr}(i\hbar\mathbb{1}) = 3i\hbar \neq 0$. Hence the canonical commutation relation can *never* be realized by finite matrices — it requires the infinite-dimensional Hilbert space.

Checks

- **Dimensions.** c has units of length, d of momentum; $cd = \hbar/2$, so $[X, P] \propto i\hbar$ as required.
- **Hermiticity.** X, P Hermitian $\Rightarrow$ $[X, P]$ is anti-Hermitian, consistent with $i\hbar\,\mathrm{diag}(1, 1, -2)$ being i times a real diagonal (Hermitian) matrix.
- **Tracelessness.** $\mathrm{tr}[X, P] = 0$ holds, as it must for any finite matrices.